REFERENCE DOCUMENT

Documentation

A practical guide for users, judges, and technical reviewers

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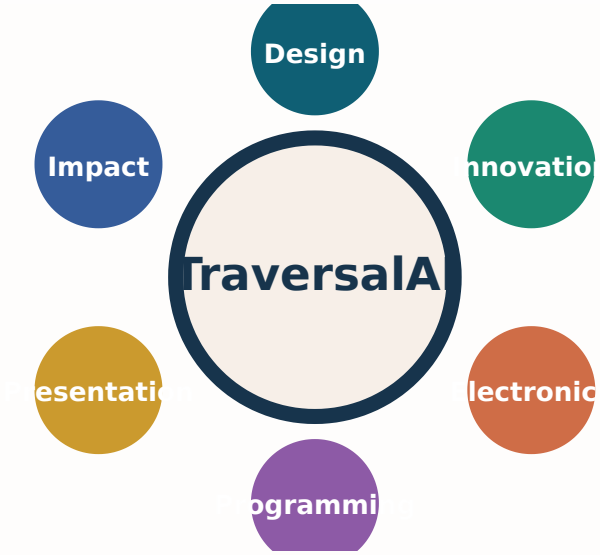
SECTION

Project Snapshot

TraversalAI is a practical prototype that combines a chat interface, a central gateway, and specialized agents in one visible system. In the challenge context it can be directed at tourism and hospitality scenarios such as smart guidance, service coordination, and multi-channel assistance. A non-technical user sees a simple interaction flow, a technical reviewer sees a testable architecture, and a judge sees a real AI + edge-computing prototype rather than a slide deck.



SECTION

Alignment With Evaluation Criteria

- Design: this document clarifies structure, boundaries, and user flow.
- Innovation: the platform unifies agents, tools, and skills inside one workflow.
- Electronics: the Raspberry Pi 5, microphone, and speaker make the AI + IoT profile explicit.
- Programming: execution passes through the gateway, saved sessions, and UI layers.
- Presentation: the content is arranged to support a short live explanation.
- Impact: the narrative keeps attention on a real tourism or hospitality problem solved through a working prototype.

SECTION

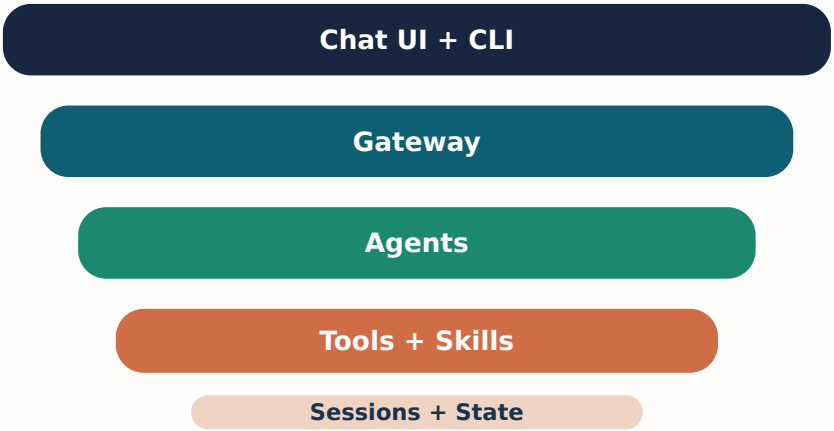
Hardware Platform

The prototype is designed to run locally on a Raspberry Pi 5 with microphone input and speaker output. That gives the project a concrete AI + IoT shape and makes it suitable for a live demonstration instead of a purely conceptual software walkthrough.

SECTION

Software Platform

The gateway is the main integration boundary. Through it, the system switches agents and models, executes tools and skills, and preserves saved chats for later continuation. That separation gives judges and technical reviewers a clear path from interface to execution.



SECTION

User Journey

- The user sends a text or voice request.
- The system selects or switches to the right agent.
- Tools and skills execute when required.
- The result appears with traceable context and saved history.

Problem **Request** **Decision** **Execution** **Outcome**

SECTION

Demo Readiness

```
1. State the problem in under 30 seconds
2. Show the device or hosted runtime
3. Run one short scenario from request to outcome
4. Expose tool or skill activity during execution
5. Close with impact and scalability
```

This document uses the guidebook rules as a reference framework and does not replace them.